

Magnesium Sulfate

- I. **Classification** – Mineral / Electrolyte
- II. **Action(s)**
 - Physiologic Electrolyte
 - Aids in cell membrane electrical stability and decreases membrane excitability
- III. **Indication(s)**
 - Ventricular Arrhythmias / Torsades de Pointes
 - Eclampsia (Pregnancy-Related Seizures)
 - Refractory asthma / bronchospasm
- IV. **Contraindications**
 - Known Hypersensitivity
 - Heart block
- V. **Precautions**
 - Caution in pulmonary edema
 - May induce hypotension
 - Caution if renal impairment
- VI. **Drug interactions**
 - Calcitriol (used in renal failure) – increased risk of hypermagnesemia
 - Narcotics and barbiturates – CNS depression
- VII. **Adult Dosing and Administration**
 - Pulseless VF / VT / Torsades de Pointes: 2 g IV / IO slow push
 - Stable Wide Complex Tachycardia Refractory to Amiodarone:
 - 2 g IV / IO mix in 100 ml NS and given over 10 minutes via pump (600 mL/hr)
 - Refractory asthma / bronchospasm: 2 g IV / IO mixed in 100 mL NS and given over 10 minutes via pump (run at 600 mL/hr)
 - Eclampsia: 4 g IV / IO mixed in 100ml NS and given over 10 minutes via Pump (run at 600 mL/hr)
 - if no IV / IO, rapidly administer 2 g IM while obtaining IV / IO, then administer remaining 2 g in 100 mL NS over 10 minutes via pump
- VIII. **Pediatric Dosing and Administration**
 - 50 mg / kg (max 2 g) IV / IO mixed in 100 ml NS and given over 10 minutes via pump (run at 600 mL/hr)
 - If Pulseless refractory VF / VT / Torsades de Pointes, administer as slow IV / IO push
- IX. **Onset**
 - Immediate
- X. **Duration**
 - 30 minutes
- XI. **Possible Adverse Effects**
 - Respiratory depression / respiratory paralysis
 - Flushing / Sweating
 - Depressed cardiac function / pulmonary edema / cardiovascular collapse